VA: Testing randomness by a circle

- This assignment is not mandatory. This assignment is for extra points during the exam. Extra points are given for this course instance only (exam or re-exam).
- To present VA, describe your code and answer questions from teaching team.
- You can choose another project for VA if (1) you did not do it for another course (2) you use skills and concepts learned during the course to a noticeable extent and (3) you emailed the idea of you project to the main teacher that is then got approved.
- You can take a look on the previous VA, but take into account three rules above.
- When the assignment has been approved you should upload the code to Studium.
- When you pass VA, teaching team will give you a (secret) code that you can use during the exam.

Important: You may collaborate with other students, but you must write and be able to explain your own code. You may not copy code neither from other students nor from the Internet. Changing variable names and similar modifications does not count as writing your own code.

This assignment were inspired by US National Institute of Standards and Technology (NIST) and their tests for randomness of binary strings. Having a source that produces random or pseudo-random binary strings is crucial for many applications including cryptographic applications such as password generation as well for simulations and games. One of such randomness test that should look partially familiar for you is described below.

The string is divided into non-overlapping binary sequences of length $2L$. For each sequence the first half represents x-coordinate of a points, and the second half represent y-coordinate (in base 2). The proportion of points falling in the circle with radius $2^L - 1$ (base 10) is calculated. Then it is tested how far the value of proportion from $\pi/4$.

In the exercises below, you will write the code to test random sequences produced in Python and also analyze and modify the test.

Let's first justify the theoretical value of proportion to be $\pi/4$. For this, the default value for the probability q of generating symbol 1 in the string should be the same as for symbol 0.

$$q = P(1) = P(0) = 0.5$$

1. Write function `proportion(L)`, where L is the length of binary sequences for each coordinate. The program returns the proportion of points lying in a circle with radius $2^L - 1$ from the set of all possible points with coordinates (x,y) representing by sequences of length L in base 2. Run the code with $L = 12$. Print both the output and its value multiplied by 4.
2. Add the second parameter being $q = 0.5$ by default. Parameter q is the probability of drawing symbol 1 (and thus not 0) in the string. Call the function `proportion_with_prob(L, q=0.5)`. No random generation is required, just theoretical value for the proportion is needed.
3. What is the complexity of your code? How long would it take to run the program with parameters $L = 24, q = 0.5$?
4. Try to improve the complexity and write the new version of the code in function `proportion_upgrade(L, q)`. Here, even the lower constant would count as improvement if you show the execution time decrease e.g. for $L = 14$. Of course, the lower complexity the better.

The interest in this problem is not only in fast calculation of the theoretical value of proportion, but also the extension to the case when q is different from 0.5. This would help to test that the sequence is random even if symbols 1 are more/less probable than symbol 0.

5. Print the results for the fixed value of $L = 14$ and for $q = 0.4$, $q = 0.5$, $q = 0.6$, $q = 0.7$. Use memoization in `proportion_upgrade(L, q)`, that is, save what is relevant to L during the first run to use for later calls of the function.

Finally, let's justify the outputs by experiments.

6. Generate a binary string using `random` (or other library) with $q = 0.7$ and length $N = 10^6$, divide it by non-overlapping sequences of length $L = 12$, where all odd sequences are x-coordinates and all even sequences are y-coordinates. Calculate how many points (x,y) are in the circle of radius $2^L - 1$.
7. Use list comprehension and/or Higher order Function in your code.
8. What is the mean value of the proportion of dots in the circle and its standard deviation? **Use multiprocessing** to answer this question.